

Session Actions Edit View Help

(kali@kali)-[~]

\$ mkdir ~/icdfa_lab_carving && cd ~/icdfa_lab_carving

(kali@kali)-[~/icdfa_lab_carving]

\$ mkdir extracted evidence hex_analysis tsk_output binwalk_output scalpel_output

(kali@kali)-[~/icdfa_lab_carving]

\$ sudo apt update

[sudo] password for kali:

Get:2 https://repo.protonvpn.com/debian stable InRelease [2,967 B]

Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]

Get:3 http://kali.download/kali kali-rolling/main amd64 Packages [21.5 MB]

Get:4 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.5 MB]

Get:5 http://kali.download/kali kali-rolling/contrib amd64 Packages [113 kB]

Get:6 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [261 kB]

Get:7 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]

Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [16.0 kB]

Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [40.

6 kB]

Fetched 75.6 MB in 1min 53s (669 kB/s)

1920 packages can be upgraded. Run 'apt list --upgradable' to see them.

Session Actions Edit View Help

(kali@kali)-[~/icdfa_lab_carving]

\$ sudo apt install sleuthkit binwalk scalpel p7zip-full xxd -y

Note, selecting '7zip' instead of 'p7zip-full'

sleuthkit is already the newest version (4.14.0+dfsg-0kali1).

sleuthkit set to manually installed.

scalpel is already the newest version (1.60+git20240110.6960eb2-2).

scalpel set to manually installed.

xxd is already the newest version (2:9.2.0858-1).

The following packages were automatically installed and are no longer required:

libpostal-data libpostal1

Use 'sudo apt autoremove' to remove them.

Upgrading:

7zip binwalk python3-binwalk

Summary:

Upgrading: 3, Installing: 0, Removing: 0, Not Upgrading: 1917

Download size: 1,738 kB

Freed space: 2,048 B

Get:1 http://http.kali.org/kali kali-rolling/main amd64 7zip amd64 26.02+dfsg-2 [1,610 k B]

Get:2 http://http.kali.org/kali kali-rolling/main amd64 python3-binwalk all 2.4.3+dfsg1-3 [122 kB]

Get:3 http://http.kali.org/kali kali-rolling/main amd64 binwalk all 2.4.3+dfsg1-3 [6,008 B]

Fetched 1,738 kB in 20s (88.2 kB/s)

(Reading database... 426373 files and directories currently installed.)

Preparing to unpack .../7zip_26.02+dfsg-2_amd64.deb...

Unpacking 7zip (26.02+dfsg-2) over (26.00+dfsg-2)...

Preparing to unpack .../python3-binwalk_2.4.3+dfsg1-3_all.deb...

Unpacking python3-binwalk (2.4.3+dfsg1-3) over (2.4.3+dfsg1-2)...

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
$ mv ~/Downloads/Ch01InChap01.dd ~/icdfa_lab_carving/evidence/
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
$ cd ~/icdfa_lab_carving/evidence
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
$ wget "https://www.dropbox.com/scl/fi/etvu2tzwrdb12w7774bwc/Ch01InChap01.dd?rlkey=<your_rlkey>&dl=1" -O Ch01InChap01.dd
--2026-09-04 13:37:06-- https://www.dropbox.com/scl/fi/etvu2tzwrdb12w7774bwc/Ch01InChap01.dd?rlkey=%3Cyour_rlkey%3E&dl=1
Resolving www.dropbox.com (www.dropbox.com)... 162.125.64.18, 2620:100:6023:18::a27d:4312
Connecting to www.dropbox.com (www.dropbox.com)|162.125.64.18|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'Ch01InChap01.dd'
```

```
Ch01InChap01.dd          [      ⇔      ] 183.97K   222KB/s   in 0.8s
```

```
2026-09-04 13:37:09 (222 KB/s) - 'Ch01InChap01.dd' saved [188385]
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
$ file Ch01InChap01.dd
```

```
Ch01InChap01.dd: HTML document, ASCII text, with very long lines (57623)
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
$
```


Session Actions Edit View Help

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_file/J_ub_law.jpg
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ ls -l
```

```
total 38960
```

```
-rw-rw-r-- 1 kali kali 36720470 Sep  4 15:00 120M.7z
-rw-rw-r-- 1 kali kali  1474560 Sep  4 13:58 Ch01InChap01.dd
-rw-rw-r-- 1 kali kali  1692150 Sep  4 15:00 J_ub_law.jpg
-rw-rw-r-- 1 kali kali      50 Sep  4 14:02 original_hashes.md5
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ file J_ub_law.jpg
```

```
file 120M.7z
```

```
J_ub_law.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, directories=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3
```

```
120M.7z: 7-zip archive data, version 0.4
```

(kali@kali)-[~/icdfa_lab_carving/evidence]

\$ md5sum Ch01InChap01.dd J_ub_law.jpg 120M.7z | tee original_hashes.md5

a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd

83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg

dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z

(kali@kali)-[~/icdfa_lab_carving/evidence]

\$ sha256sum Ch01InChap01.dd J_ub_law.jpg 120M.7z | tee original_hashes.sha256

3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 Ch01InChap01.dd

238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law.jpg

2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce 120M.7z

(kali@kali)-[~/icdfa_lab_carving/evidence]

\$ █

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ xxd J_ub_law.jpg | head -5
```

```
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013  ....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201  ....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100  ....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400  ....
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ xxd J_ub_law.jpg | tail -5
```

```
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80  )0.J.Y..(.QZ.v:.
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45  ....y....S..._E
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1  .*.-$....i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565  ..RT.8..o..Z.T.e
0019d1f0: d351 9ebf ffd9  .Q....
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]  
$ mkdir -p ../hex_analysis
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]  
$ xxd -p J_ub_law.jpg > ../hex_analysis/J_ub_law_plain.hex
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]  
$ head -c 200 ../hex_analysis/J_ub_law_plain.hex  
ffd8ffe117f045786966000049492a00080000001000000010300010000000  
40130000001010300010000000d00c00000201030003000000ce0000000601  
03000100000000200000000f01020012000000d40000001001020009000000  
e6000000120103000
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]  
$
```



```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ xxd -r -p ../hex_analysis/J_ub_law_plain.hex ../hex_analysis/J_ub_law_reconstructed.jpg
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ file ../hex_analysis/J_ub_law_reconstructed.jpg
```

```
../hex_analysis/J_ub_law_reconstructed.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components 3
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ md5sum J_ub_law.jpg ../hex_analysis/J_ub_law_reconstructed.jpg
```

```
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
```

```
83a360ac7f7e0ca318e5bfe39f95f137 ../hex_analysis/J_ub_law_reconstructed.jpg
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ sha256sum J_ub_law.jpg ../hex_analysis/J_ub_law_reconstructed.jpg
```

```
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law.jpg
```

```
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 ../hex_analysis/J_ub_law_reconstructed.jpg
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ █
```



```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ strings J_ub_law.jpg | less
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ strings J_ub_law.jpg | grep -iE "nikon|canon|sony|adobe|photoshop|software|camera|copyright|author|date"
```

```
NIKON CORPORATION
```

```
NIKON D4
```

```
Adobe Photoshop 21.1 (Macintosh)
```

```
Adobe_CM
```

```
Adobe
```

```
"Photoshop 3.0
```

```
Adobe_CM
```

```
Adobe
```

```
http://ns.adobe.com/xap/1.0/
```

```
" id="W5M0MpCehiHzreSzNTczkc9d"?> <x:xmpmeta xmlns:x="adobe:ns:meta/" x:xmptk="Adobe XMP Core 6.0-c002 79.164352, 2020/01/30-15:50:38" >
```

```
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" > <rdf:Description rdf:about="" xmlns:aux="http://ns.adobe.com/exif/1.0/aux/" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/" xmlns:xmp="http://ns.adobe.com/xap/1.0/" xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/" xmlns:stRef="http://ns.adobe.com/xap/1.0/sType/ResourceRef#" xmlns:stEvt="http://ns.adobe.com/xap/1.0/sType/ResourceEvent#" xmlns:crs="http://ns.adobe.com/camera-raw-settings/1.0/" xmlns:lr="http://ns.adobe.com/lightroom/1.0/" aux:ApproximateFocusDistance="4294967295/1" aux:ImageNumber="115422" aux:Lens="17.0-35.0 mm f/2.8" aux:LensID="99"
```

```
" aux:LensInfo="170/10 350/10 28/10 28/10" aux:SerialNumber="2027750" dc:format="image/jpeg" photoshop:DateCreated="2013-10-08" photoshop:LegacyIPTCDigest="5E379AE71DB032222FA2DE651E47F460" photoshop:ColorMode="3" photoshop:ICCProfile="Adobe RGB (1998)" xmp:CreateDate="2013-10-08T18:56:21" xmp:CreatorTool="NIKON D4 Ver.1.01" xmp:Label="Approved" xmp:MetadataDate="2020-08-20T10:20:05-04:00" xmp:ModifyDate="2020-08-20T10:20:05-04:00" xmp:Rating="0" xmpMM:DocumentID="adobe:docid:phot
```

Session Actions Edit View Help

(kali@kali)-[~/icdfa_lab_carving/evidence]

\$ xxd J_ub_lab.jpg | head -40

```
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 .....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201 .....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100 .....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400 .....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201 .....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100 .....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00 .....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801 .....(
00000090: 0300 0100 0000 0200 0000 3101 0200 2100 .....1...!.
000000a0: 0000 ff00 0000 3201 0200 1400 0000 2001 .....2.....
000000b0: 0000 9882 0200 0c00 0000 3401 0000 9e9c .....4.....
000000c0: 0100 5200 0000 4001 0000 6987 0400 0100 ..R...@...i....
000000d0: 0000 9401 0000 4404 0000 0800 0800 0800 .....D.....
000000e0: 4e49 4b4f 4e20 434f 5250 4f52 4154 494f NIKON CORPORATIO
000000f0: 4e00 4e49 4b4f 4e20 4434 00c0 c62d 0010 N.NIKON D4...-..
00000100: 2700 00c0 c62d 0010 2700 0041 646f 6265 '....-..'..Adobe
00000110: 2050 686f 746f 7368 6f70 2032 312e 3120 Photoshop 21.1
00000120: 284d 6163 696e 746f 7368 2900 3230 3230 (Macintosh).2020
00000130: 3a30 383a 3230 2031 303a 3230 3a30 3500 :08:20 10:20:05.
00000140: 484f 5741 5244 204b 4f52 4e00 3200 3000 HOWARD KORN.2.0.
00000150: 3100 3300 2c00 2000 6300 6100 6d00 7000 1.3.,. .c.a.m.p.
00000160: 7500 7300 2c00 2000 4d00 6500 7200 7200 u.s.,. .M.e.r.r.
00000170: 6900 6300 6b00 2000 5300 6300 6800 6f00 i.c.k. .S.c.h.o.
00000180: 6f00 6c00 2000 6f00 6600 2000 4200 7500 o.l. .o.f. .B.u.
00000190: 7300 6900 6e00 6500 7300 7300 0000 0000 s.i.n.e.s.s.....
000001a0: 2900 9a82 0500 0100 0000 8603 0000 9d82 ).....
000001b0: 0500 0100 0000 8e03 0000 2288 0300 0100 .....".
000001c0: 0000 0100 0000 2788 0300 0100 0000 c800 .....'.
000001d0: 0000 3088 0300 0100 0000 0200 0000 0000 ..0
```


Session Actions Edit View Help

(kali@kali)-[~/icdfa_lab_carving/evidence]

\$ which exiftool || sudo apt install libimage-exiftool-perl -y
/usr/bin/exiftool

(kali@kali)-[~/icdfa_lab_carving/evidence]

\$ exiftool J_ub_law.jpg

ExifTool Version Number	: 13.50
File Name	: J_ub_law.jpg
Directory	: .
File Size	: 1692 kB
File Modification Date/Time	: 2026:09:04 15:00:15-04:00
File Access Date/Time	: 2026:09:04 15:01:17-04:00
File Inode Change Date/Time	: 2026:09:04 15:00:15-04:00
File Permissions	: -rw-rw-r--
File Type	: JPEG
File Type Extension	: jpg
MIME Type	: image/jpeg
Exif Byte Order	: Little-endian (Intel, II)
Photometric Interpretation	: RGB
Make	: NIKON CORPORATION
Camera Model Name	: NIKON D4
Orientation	: Horizontal (normal)
Samples Per Pixel	: 3
X Resolution	: 300
Y Resolution	: 300
Resolution Unit	: inches
Software	: Adobe Photoshop 21.1 (Macintosh)
Modify Date	: 2020:08:20 10:20:05
Copyright	: HOWARD KORN
XP Keywords	: 2013, campus, Merrick School of Business
Exposure Time	: 8
F Number	: 11.0

Session Actions Edit View Help

Exposure Program	: Manual
ISO	: 200
Sensitivity Type	: Recommended Exposure Index
Exif Version	: 0221
Date/Time Original	: 2013:10:08 18:56:21
Create Date	: 2013:10:08 18:56:21
Shutter Speed Value	: 8
Aperture Value	: 11.0
Exposure Compensation	: 0
Max Aperture Value	: 2.8
Metering Mode	: Multi-segment
Light Source	: Unknown
Flash	: No Flash
Focal Length	: 20.0 mm
Sub Sec Time Original	: 80
Color Space	: Uncalibrated
Exif Image Width	: 1920
Exif Image Height	: 1080
Focal Plane X Resolution	: 1368.888885
Focal Plane Y Resolution	: 1368.888885
Focal Plane Resolution Unit	: cm
Sensing Method	: One-chip color area
File Source	: Digital Camera
Scene Type	: Directly photographed
CFA Pattern	: [Red,Green][Green,Blue]
Custom Rendered	: Normal
Exposure Mode	: Manual
White Balance	: Auto
Digital Zoom Ratio	: 1
Focal Length In 35mm Format	: 20 mm
Scene Capture Type	: Standard
Gain Control	: None

(kali@kali)-[~/icdfa_lab_carving/evidence]

\$ img_stat Ch01InChap01.dd

IMAGE FILE INFORMATION

Image Type: raw

Size in bytes: 1474560

Sector size: 512

(kali@kali)-[~/icdfa_lab_carving/evidence]

\$ mmls Ch01InChap01.dd

(kali@kali)-[~/icdfa_lab_carving/evidence]

\$ fsstat Ch01InChap01.dd

FILE SYSTEM INFORMATION

File System Type: FAT12

OEM Name: , 'k0nIHC

Volume ID: 0x0

Volume Label (Boot Sector):

Volume Label (Root Directory):

File System Type Label: ♦3♦♦M|

Sectors before file system: 0

File System Layout (in sectors)

Total Range: 0 - 2879

* Reserved: 0 - 0

** Boot Sector: 0

* FAT 0: 1 - 9

```
(kali@kali)-[~/icdfa_lab_carving/evidence]  
$ fls -r Ch01InChap01.dd  
r/r 5: Client Info.mdb  
r/r * 8: Billing Letter.doc  
r/r * 11: confirmation.txt  
r/r 13: Income.xls  
r/r * 15: letter1.txt  
r/r * 17: Regrets.doc  
v/v 45779: $MBR  
v/v 45780: $FAT1  
v/v 45781: $FAT2  
v/v 45782: $OrphanFiles
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]  
$ █
```



```
(kali@kali)-[~/icdfa_lab_carving/evidence]  
$ icat Ch01InChap01.dd 11 > ../tsk_output/recovered_confirmation.txt
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]  
$ cat ../tsk_output/recovered_confirmation.txt
```

Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.

ftp.acmeserver.com

login id: laura.roper
password: 342rroi9

Thank you for your business

George

```
(kali@kali)-[~/icdfa_lab_carving/evidence]  
$ file ../tsk_output/recovered_confirmation.txt  
../tsk_output/recovered_confirmation.txt: ASCII text, with CRLF line terminators
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]  
$ icat Ch01InChap01.dd 8 > ../tsk_output/recovered_billing_letter.doc
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ file ../tsk_output/recovered_billing_letter.doc
```

```
../tsk_output/recovered_billing_letter.doc: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Title: 13 October 2003, Author: Amelia Phillips, Template: Normal.dot, Last Saved By: Amelia Phillips, Revision Number: 2, Name of Creating Application: Microsoft Word 9.0, Total Editing Time: 06:00, Create Time/Date: Sat Nov 23 03:06:00 2002, Last Saved Time/Date: Fri Dec 9 14:50:00 2005, Number of Pages: 1, Number of Words: 118, Number of Characters: 678, Security: 0
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ blkcat Ch01InChap01.dd 33 > ../tsk_output/recovered_block33.bin
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ file ../tsk_output/recovered_block33.bin
```

```
../tsk_output/recovered_block33.bin: Microsoft Access Database
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ xxd ../tsk_output/recovered_block33.bin | head -10
```

```
00000000: 0001 0000 5374 616e 6461 7264 204a 6574 ....Standard Jet
00000010: 2044 4200 0000 0000 b56e 0362 6009 c255 DB.....n.b`..U
00000020: e9a9 6772 403f 009c 7e9f 90ff 859a 31c5 ..gr@?..~.....1.
00000030: 79ba ed30 bcd5 cc9d 63d9 edc7 9f46 fb8a y..0....c....F..
00000040: bc4e 86fb ec37 5d44 9cfa c65e 28e6 13b6 .N...7]D...^(...
00000050: 8a60 5494 7b36 b67d dfb1 77f4 1343 cfaf .`T.{6.}..w..C..
00000060: b133 3461 795b 92b5 7c2a 05f1 7c99 011b .34ay[..|*..|...
00000070: 98fd 124f 4a94 6c3e 6026 5f95 f8d0 8924 ...OJ.l>`&_....$
00000080: 8567 c61f 2744 d2ee cf65 edff 07c7 46a1 .g..`D...e....F.
00000090: 7816 0ced e92d 0000 0000 0000 0000 0000 x....~.....
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ cd ~/icdfa_lab_carving/tsk_output
```

```
(kali@kali)-[~/icdfa_lab_carving/tsk_output]
```

```
$ sha256sum recovered_confirmation.txt recovered_billing_letter.doc recovered_bloc
```

```
k33.bin | tee recovered_hashes.sha256
```

```
51b981e09b72739fac67c2d0124a2603cae8c35e075f5e2f66fbad03d53afcc0 recovered_confirma  
tion.txt
```

```
7eefeb4187bb551f366da33bcab5b7224f905bb98f37797c28f183f72aec2e99 recovered_billing_  
letter.doc
```

```
d7677a17534ac996430996b9b5ab41e68a0c4a7f0aafde3ff1caa4c45702dad6 recovered_block33.  
bin
```



```
(kali㉿kali)-[~/icdfa_lab_carving/tsk_output]
```

```
$ cd ~/icdfa_lab_carving/evidence
```

```
(kali㉿kali)-[~/icdfa_lab_carving/evidence]
```

```
$ icat Ch01InChap01.dd 15 > ../tsk_output/recovered_letter1.txt
```

```
(kali㉿kali)-[~/icdfa_lab_carving/evidence]
```

```
$ cat ../tsk_output/recovered_letter1.txt
```

Earl,

We need to meet on the 18th of August to confirm the work I am doing for you. Please contact me ASAP.

George

```
(kali㉿kali)-[~/icdfa_lab_carving/evidence]
```

```
$ file ../tsk_output/recovered_letter1.txt
```

```
../tsk_output/recovered_letter1.txt: ASCII text, with CRLF line terminators
```

```
(kali㉿kali)-[~/icdfa_lab_carving/evidence]
```

```
$ sha256sum ../tsk_output/recovered_letter1.txt | tee -a ../tsk_output/recovered_hashes.sha256
```

```
341065f2c3c2168e246aa9a346439dfa4e6e4c55de4b1b8ecc78e4b83206545f  ../tsk_output/recovered_letter1.txt
```

```
(kali㉿kali)-[~/icdfa_lab_carving/evidence]
```

```
$
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]  
$ binwalk J_ub_law.jpg
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	JPEG image data, EXIF standard
12	0xC	TIFF image data, little-endian offset of first image d
directory: 8		
26844	0x68DC	Copyright string: "Copyright 1999 Adobe Systems Incorp
orated"		

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
$ sudo apt install binwalk-full -y 2>/dev/null || pip install python-lzo --break-s
system-packages
[sudo] password for kali:
Defaulting to user installation because normal site-packages is not writeable
Collecting python-lzo
  Downloading python-lzo-1.15.tar.gz (17 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: python-lzo
  Building wheel for python-lzo (pyproject.toml) ... error
error: subprocess-exited-with-error

  Building wheel for python-lzo (pyproject.toml) did not run successfully.
  exit code: 1
  > [7 lines of output]
    /tmp/pip-build-env-x2jflqt9/overlay/local/lib/python3.13/dist-packages/setupto
ols/_distutils/dist.py:318: UserWarning: Unknown distribution option: 'tests_require
'
        warnings.warn(msg)
    lzomodule.c:38:10: fatal error: lzo/lzo1.h: No such file or directory
       38 | #include <lzo/lzo1.h>
          |          ^~~~~~
    compilation terminated.
    error: Command '['x86_64-linux-gnu-gcc', '-fno-strict-overflow', '-Wsign-compa
re', '-DNDEBUG', '-g', '-O2', '-Wall', '-fPIC', '-I/usr/include', '-I/usr/include/py
thon3.13', '-c', 'lzomodule.c', '-o', 'build/temp.linux-x86_64-cpython-313/lzomodule
.o']' returned non-zero exit status 1.
    [end of output]

note: This error originates from a subprocess, and is likely not a problem with pi
```




File Edit View Go Help



```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ mkdir -p ../binwalk_output
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ dd if=J_ub_law.jpg of=../binwalk_output/extracted_thumbnail.jpg bs=1 skip=1198 c  
ount=4934
```

```
4934+0 records in
```

```
4934+0 records out
```

```
4934 bytes (4.9 kB, 4.8 KiB) copied, 0.0252108 s, 196 kB/s
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ file ../binwalk_output/extracted_thumbnail.jpg
```

```
../binwalk_output/extracted_thumbnail.jpg: JPEG image data, baseline, precision 8, 1  
60x90, components 3
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ xdg-open ../binwalk_output/extracted_thumbnail.jpg
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ sha256sum ../binwalk_output/extracted_thumbnail.jpg | tee ../binwalk_output/thum  
bnail_hash.sha256
```

```
f47359e0aca4b23c876ac78888819e0c5a58e71e53c0a1a0f9005115a20c4c9d  ../binwalk_output/  
extracted_thumbnail.jpg
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ mkdir -p ../extracted_usb
```

```
(kali@kali)-[~/icdfa_lab_carving/evidence]
```

```
$ 7z x 120M.7z -o../extracted_usb
```

```
7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25  
64-bit locale=en_US.UTF-8 Threads:128 OPEN_MAX:4096, ASM
```

```
Scanning the drive for archives:  
1 file, 36720470 bytes (36 MiB)
```

```
Extracting archive: 120M.7z
```

```
--
```

```
Path = 120M.7z
```

```
Type = 7z
```

```
Physical Size = 36720470
```

```
Headers Size = 214
```

```
Method = LZMA2:24
```

```
Solid = +
```

```
Blocks = 1
```

```
Everything is Ok
```

```
Folders: 1
```

```
Files: 2
```

```
Size: 124782229
```

```
Compressed: 36720470
```



```
(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]
```

```
$ file usb_fat_carving.001
```

```
usb_fat_carving.001: DOS/MBR boot sector MS-MBR Windows 7 english at offset 0x163 "Invalid partition table" at offset 0x17b "Error loading operating system" at offset 0x19a "Missing operating system", disk signature 0xa1159a00; partition 1 : ID=0xe, active, start-CHS (0x0,2,3), end-CHS (0xe,254,63), startsector 128, 243584 sectors
```

```
(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]
```

```
$ md5sum usb_fat_carving.001 | tee usb_image_original.md5
```

```
ba4a1d0ba49f4a6667b00a3b3e85e604  usb_fat_carving.001
```

```
(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]
```

```
$ sha256sum usb_fat_carving.001 | tee usb_image_original.sha256
```

```
9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9  usb_fat_carving.001
```

```
(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]
```

```
$ mmls usb_fat_carving.001
```

```
DOS Partition Table
```

```
Offset Sector: 0
```

```
Units are in 512-byte sectors
```

	Slot	Start	End	Length	Description
000:	Meta	0000000000	0000000000	0000000001	Primary Table (#0)
001:	_____	0000000000	0000000127	0000000128	Unallocated
002:	000:000	0000000128	0000243711	0000243584	Win95 FAT16 (0x0e)

```
(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]
```

```
$
```

(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]

\$ fsstat -o 128 usb_fat_carving.001 | tee ../../tsk_output/usb_fsstat.txt

FILE SYSTEM INFORMATION

File System Type: FAT16

OEM Name: MSDOS5.0

Volume ID: 0x94105672

Volume Label (Boot Sector): NO NAME

Volume Label (Root Directory): USB

File System Type Label: FAT16

Sectors before file system: 128

File System Layout (in sectors)

Total Range: 0 - 243583

* Reserved: 0 - 3

** Boot Sector: 0

* FAT 0: 4 - 241

* FAT 1: 242 - 479

* Data Area: 480 - 243583

** Root Directory: 480 - 511

** Cluster Area: 512 - 243583

METADATA INFORMATION

Range: 2 - 3889670

Root Directory: 2

CONTENT INFORMATION

Sector Size: 512

Session Actions Edit View Help

CONTENT INFORMATION

Sector Size: 512

Cluster Size: 2048

Total Cluster Range: 2 - 60769

FAT CONTENTS (in sectors)

512-515 (4) → EOF

516-519 (4) → EOF

520-523 (4) → EOF

70976-71211 (236) → EOF

71212-71215 (4) → EOF

```
(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]
$ fls -o 128 -r usb_fat_carving.001 | tee ../../tsk_output/usb_fls.txt
r/r 3: USB (Volume Label Entry)
d/d 6: System Volume Information
+ r/r 519: WPSettings.dat
r/r * 7: _est
r/r 10: .dropbox.device
d/d * 13: old_File_Carving_files
+ r/r * 711: CarvedImage.jpg
+ r/r * 714: File_carving.docx
+ r/r * 717: File_Carving.pptx
+ r/r * 720: i-love-cats-cute-cats.jpg
r/r * 15: B_ub_poe4.bmp
r/r * 18: B_zoom-eubie-mono.bmp
r/r * 21: Ballardlab8.java
r/r * 24: brittLab10.java
r/r * 27: D0_example.doc
r/r * 30: D0_example2.doc
r/r * 33: G_BuiltForThis.gif
r/r * 35: G_zoom-sc.gif
r/r * 37: H_Form.html
r/r * 39: H_hello.html
r/r * 41: J_ub_law.jpg
r/r * 44: J_ub_night.jpg
r/r * 47: nps-2008-jean_outlook.pst
r/r * 50: P_CAS-zoom-6.png
r/r * 53: P_MSB_1_zoom.png
r/r * 57: pd_Evidence_search_techniques.pdf
r/r * 61: pd_Forensic_Report_Template.pdf
r/r * 64: pp_Number_Systems.pptx
r/r * 67: pp_one_page.pptx
r/r 69: readme.docx
```



```
r/r 69: readme.docx
r/r 70: readme.txt
r/r * 71:      _tf_1.rtf
r/r * 74:      T_Eubie-iphone-5-8.tiff
r/r * 78:      T_youknowus-iphone-5-8.tiff
r/r * 81:      W_axloadcomplete.wav
r/r * 84:      W_speaker_test_sound.wav
r/r * 86:      Z_file.zip
r/r * 88:      江莎莎行踪轨迹.doc
r/r * 91:      nps-2008-jean_outlook.pst
v/v 3889667:   $MBR
v/v 3889668:   $FAT1
v/v 3889669:   $FAT2
V/V 3889670:   $OrphanFiles
```

```
(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]
$ █
```

```
(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]
$ icat -o 128 usb_fat_carving.001 41 > ../../tsk_output/recovered_J_ub_law_via_ica
t.jpg
```

```
(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]
$ file ../../tsk_output/recovered_J_ub_law_via_icat.jpg
../../tsk_output/recovered_J_ub_law_via_icat.jpg: JPEG image data, Exif standard: [T
IFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206, Phot
ometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientati
on=upper-left], baseline, precision 8, 1920x1080, components 3
```

```
(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]
$ sha256sum ../../tsk_output/recovered_J_ub_law_via_icat.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  ../../tsk_output/r
ecovered_J_ub_law_via_icat.jpg
```

```
(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]
$ sudo nano /etc/scalpel/scalpel.conf
[sudo] password for kali:
```

```
(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]
$
```

Session Actions Edit View Help

GNU nano 8.7.1 /etc/scalpel/scalpel.conf *

```
# beginning with the string "FOREMOST" are carved and no file extensions
# are used. No footer is defined and the max carve size is 1000 bytes.
```

```
#      NONE      y      1000      FOREMOST
```

GRAPHICS FILES

AOL ART files

```
#      art      y      150000  \x4a\x47\x04\x0e      \xcf\xcf\xcb
#      art      y      150000  \x4a\x47\x03\x0e      \xd0\xcb\x00\x00
```

GIF and JPG files (very common)

```
#      gif      y      5000000      \x47\x49\x46\x38\x37\x61      \x00\x3b
#      gif      y      5000000      \x47\x49\x46\x38\x39\x61      \x00\x3b
#      jpg      y      5242880      \xff\xd8\xff???Exif      \xff\xd9
#      jpg      y      5242880      \xff\xd8\xff???JFIF      \xff\xd9
```

PNG

```
#      png      y      20000000      \x50\x4e\x47?      \xff\xfc\xfd\xfe
```

```
# BMP (used by MSWindows, use only if you have reason to think there are
# BMP files worth digging for. This often kicks back a lot of false
# positives
```

```
(kali@kali)-[~/icdfa_lab_carving/extracted_usb/120M]
$ cd ~/icdfa_lab_carving
scalpel -c /etc/scalpel/scalpel.conf -o scalpel_output extracted_usb/120M/usb_fat_carving.001
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/kali/icdfa_lab_carving/extracted_usb/120M/usb_fat_carving.001"

Image file pass 1/2.
extracted_usb/120M/usb_fat_carving.001: 100.0% [*****] 119.0 MB 00:00 ETA
Allocating work queues ...
Work queues allocation complete. Building carve lists ...
Carve lists built. Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" →
8 files
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" →
9 files
Carving files from image.
Image file pass 2/2.
extracted_usb/120M/usb_fat_carving.001: 100.0% [*****] 119.0 MB 00:00 ETA
Processing of image file complete. Cleaning up ...
Done.
Scalpel is done, files carved = 17, elapsed = 1 seconds.

(kali@kali)-[~/icdfa_lab_carving]
$
```



```
(kali㉿kali)-[~/icdfa_lab_carving]
$ cd ~/icdfa_lab_carving/scalpel_output
cat audit.txt
```

```
Scalpel version 1.60 audit file
Started at Sat Sep  5 05:18:29 2026
Command line:
scalpel -c /etc/scalpel/scalpel.conf -o scalpel_output extracted_usb/120M/usb_fat_carving.001
```

```
Output directory: /home/kali/icdfa_lab_carving/scalpel_output
Configuration file: /etc/scalpel/scalpel.conf
```

```
Opening target "/home/kali/icdfa_lab_carving/extracted_usb/120M/usb_fat_carving.001"
```

The following files were carved:

File	Start	Chop	Length	Extracted From
00000010.jpg	3796992	NO	3148500	usb_fat_carving.001
00000009.jpg	425822	NO	4875802	usb_fat_carving.001
00000008.jpg	335872	NO	4965752	usb_fat_carving.001
00000002.jpg	6938624	NO	6868	usb_fat_carving.001
00000001.jpg	5285888	NO	1659604	usb_fat_carving.001
00000000.jpg	3897344	NO	3048148	usb_fat_carving.001
00000004.jpg	15427584	NO	4223822	usb_fat_carving.001

```
(kali@kali)-[~/icdfa_lab_carving/scalpel_output]
$ ls -la
total 20
drwxrwxr-x 4 kali kali 4096 Sep  5 05:18 .
drwxrwxr-x 9 kali kali 4096 Sep  5 05:01 ..
-rw-rw-r-- 1 kali kali 1648 Sep  5 05:18 audit.txt
drwxrwxr-x 2 kali kali 4096 Sep  5 05:18 jpg-0-0
drwxrwxr-x 2 kali kali 4096 Sep  5 05:18 jpg-1-0
```

```
(kali@kali)-[~/icdfa_lab_carving/scalpel_output]
$ find . -type f -name "*.jpg"
./jpg-0-0/00000000.jpg
./jpg-0-0/00000002.jpg
./jpg-0-0/00000001.jpg
./jpg-0-0/00000006.jpg
./jpg-0-0/00000007.jpg
./jpg-0-0/00000004.jpg
./jpg-0-0/00000005.jpg
./jpg-0-0/00000003.jpg
./jpg-1-0/00000012.jpg
./jpg-1-0/00000010.jpg
./jpg-1-0/00000008.jpg
./jpg-1-0/00000014.jpg
./jpg-1-0/00000016.jpg
./jpg-1-0/00000009.jpg
./jpg-1-0/00000015.jpg
./jpg-1-0/00000013.jpg
./jpg-1-0/00000011.jpg
```

```
(kali@kali)-[~/icdfa_lab_carving/scalpel_output]
$ sha256sum $(find . -type f -name "*.jpg") | tee recovered_jpg_hashes.sha256
df36e6bcb12c782ab67a6662b5d8d306383cfe67ca050485a0f97d2db1aff293 ./jpg-0-0/00000000
.jpg
15e20d485aa30904f7b665c2ff297e8a9b01d0fc30b8dd0b4ffe9a47f4376a18 ./jpg-0-0/00000002
.jpg
a3f54f4e312899bcd60f31fe99a3c392b3b2c46555c993f1d22581980073be33 ./jpg-0-0/00000001
.jpg
0ff923813d2911727c7b72362f0d0990755a86f2bc6ca927ee8fcd0616a90573 ./jpg-0-0/00000006
.jpg
9b8a2249a0ecbd7abfdac4d98cd031f61c4e740312a09ba3f2d132d1c04bae62 ./jpg-0-0/00000007
.jpg
19b2e1bf7cdb40192d73157a1c70af5cfd89959125416c0a21a6afcd2ffd384e ./jpg-0-0/00000004
.jpg
8e49a4423aa2415b530b43b5d8ab9dbe72e1f6578e65dddd6e4202938be380e1 ./jpg-0-0/00000005
.jpg
a8c87f31f3876d99110dd91a071697a424ac7f02aca4607941f7dacc26bc47f ./jpg-0-0/00000003
.jpg
371807a9173648494a5ec4c3d63739ae6adeaab3646c15e29421eb64d46fbee4 ./jpg-1-0/00000012
.jpg
7c127f3b90c3a0052cc019392ef4bb5ec17d7ec8a39fdd17d01ccd6a58be49ab ./jpg-1-0/00000010
.jpg
3eeeb18dbaaf50a6ae3cbb7630bd7fe426b317eb79fc951f3a6ceb601f8639d1 ./jpg-1-0/00000008
.jpg
eb939a9b48da622896cf8577e9ccbd1c38a4b1686f09b3512282942d0c826411 ./jpg-1-0/00000014
.jpg
5b7a35a6b0bd67deaa1dbd92ee6b31950ace908c44f5124639d180a5e40e8567 ./jpg-1-0/00000016
.jpg
5cf67210983c08d8800e46cbd771b22b4a6bb0cccf9d96130b2106d6845094b06 ./jpg-1-0/00000009
.jpg
3d31fb0df55a4558446f8577cb7943a7dbd9fe44f782fdc8d64b13c4120d89d4 ./jpg-1-0/00000015
.jpg
```



```
(kali@kali)-[~/icdfa_lab_carving/scalpel_output]
```

```
$ file $(find . -type f -name "*.jpg" | head -5)
```

```
./jpg-0-0/00000000.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian,
dentries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionu
nit=2, software=Adobe Photoshop 21.1 (Macintosh), datetime=2020:04:03 14:35:44], bas
eline, precision 8, 1920x1080, components 3
```

```
./jpg-0-0/00000002.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian
, dentries=12, width=16550, height=9313, bps=0, PhotometricInterpretation=RGB, ori
entation=upper-left]
```

```
./jpg-0-0/00000001.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian
, dentries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionu
nit=2, software=Adobe Photoshop 21.1 (Macintosh), datetime=2020:04:03 11:54:50], bas
eline, precision 8, 1920x1080, components 3
```

```
./jpg-0-0/00000006.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian
, dentries=7, orientation=upper-left, xresolution=98, yresolution=106, resolutionu
nit=2, software=Adobe Photoshop 21.1 (Macintosh), datetime=2020:04:23 15:24:33], bas
eline, precision 8, 1920x1080, components 3
```

```
./jpg-0-0/00000007.jpg: JPEG image data, Exif standard: [TIFF image data, big-endian
, dentries=12, width=1920, height=1080, bps=0, PhotometricInterpretation=RGB, orie
ntation=upper-left], baseline, precision 8, 1920x1080, components 3
```



```
(kali@kali) - [~/icdfa_lab_carving/scalpel_output]  
$ xdg-open ./jpg-0-0/0000000000.jpg
```

```
(kali@kali) - [~/icdfa_lab_carving/scalpel_output]  
$ xdg-open ./jpg-0-0/0000000001.jpg
```

```
(kali@kali) - [~/icdfa_lab_carving/scalpel_output]  
$ █
```



File Edit View Go Help





File Edit View Go Help




```
(kali@kali)-[~/icdfa_lab_carving/scalpel_output]
$ cat audit.txt
```

```
Scalpel version 1.60 audit file
Started at Sat Sep  5 05:18:29 2026
Command line:
scalpel -c /etc/scalpel/scalpel.conf -o scalpel_output extracted_usb/120M/usb_fat_carving.001
```

```
Output directory: /home/kali/icdfa_lab_carving/scalpel_output
Configuration file: /etc/scalpel/scalpel.conf
```

```
Opening target "/home/kali/icdfa_lab_carving/extracted_usb/120M/usb_fat_carving.001"
```

```
The following files were carved:
```

File	Start	Chop	Length	Extracted From
00000010.jpg	3796992	NO	3148500	usb_fat_carving.001
00000009.jpg	425822	NO	4875802	usb_fat_carving.001
00000008.jpg	335872	NO	4965752	usb_fat_carving.001
00000002.jpg	6938624	NO	6868	usb_fat_carving.001
00000001.jpg	5285888	NO	1659604	usb_fat_carving.001
00000000.jpg	3897344	NO	3048148	usb_fat_carving.001
00000004.jpg	15427584	NO	4223822	usb_fat_carving.001
00000003.jpg	12881920	NO	4256310	usb_fat_carving.001


```
(kali@kali)-[~/icdfa_lab_carving/scalpel_output]
$ find . -type f -name "*.jpg" | xargs sha256sum | tee recovered_jpg_hashes.sha256
```

```
df36e6bcb12c782ab67a6662b5d8d306383cfe67ca050485a0f97d2db1aff293 ./jpg-0-0/00000000
.jpg
15e20d485aa30904f7b665c2ff297e8a9b01d0fc30b8dd0b4ffe9a47f4376a18 ./jpg-0-0/00000002
.jpg
a3f54f4e312899bcd60f31fe99a3c392b3b2c46555c993f1d22581980073be33 ./jpg-0-0/00000001
.jpg
0ff923813d2911727c7b72362f0d0990755a86f2bc6ca927ee8fcd0616a90573 ./jpg-0-0/00000006
.jpg
9b8a2249a0ecbd7abfdac4d98cd031f61c4e740312a09ba3f2d132d1c04bae62 ./jpg-0-0/00000007
.jpg
19b2e1bf7cdb40192d73157a1c70af5cfd89959125416c0a21a6afcd2ffd384e ./jpg-0-0/00000004
.jpg
8e49a4423aa2415b530b43b5d8ab9dbe72e1f6578e65dddd6e4202938be380e1 ./jpg-0-0/00000005
.jpg
a8c87f31f3876d99110dd91a071697a424ac7f02aca4607941f7dacc26bc47f ./jpg-0-0/00000003
.jpg
371807a9173648494a5ec4c3d63739ae6adeaab3646c15e29421eb64d46fbee4 ./jpg-1-0/00000012
.jpg
7c127f3b90c3a0052cc019392ef4bb5ec17d7ec8a39fdd17d01ccd6a58be49ab ./jpg-1-0/00000010
.jpg
3eeeb18dbaaf50a6ae3cbb7630bd7fe426b317eb79fc951f3a6ceb601f8639d1 ./jpg-1-0/00000008
.jpg
eb939a9b48da622896cf8577e9ccbd1c38a4b1686f09b3512282942d0c826411 ./jpg-1-0/00000014
.jpg
5b7a35a6b0bd67deaa1dbd92ee6b31950ace908c44f5124639d180a5e40e8567 ./jpg-1-0/00000016
.jpg
5cf67210983c08d8800e46cbd771b22b4a6bb0cccf96130b2106d6845094b06 ./jpg-1-0/00000009
.jpg
```

(kali@kali)-[~/icdfa_lab_carving/scalpel_output]

\$ sha256sum ~/icdfa_lab_carving/evidence/J_ub_law.jpg

238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 /home/kali/icdfa_lab_carving/evidence/J_ub_law.jpg

(kali@kali)-[~/icdfa_lab_carving/scalpel_output]

\$ cat recovered_jpg_hashes.sha256

df36e6bcb12c782ab67a6662b5d8d306383cfe67ca050485a0f97d2db1aff293 ./jpg-0-0/00000000
.jpg

15e20d485aa30904f7b665c2ff297e8a9b01d0fc30b8dd0b4ffe9a47f4376a18 ./jpg-0-0/00000002
.jpg

a3f54f4e312899bcd60f31fe99a3c392b3b2c46555c993f1d22581980073be33 ./jpg-0-0/00000001
.jpg

0ff923813d2911727c7b72362f0d0990755a86f2bc6ca927ee8fcd0616a90573 ./jpg-0-0/00000006
.jpg

9b8a2249a0ecbd7abfdac4d98cd031f61c4e740312a09ba3f2d132d1c04bae62 ./jpg-0-0/00000007
.jpg

19b2e1bf7cdb40192d73157a1c70af5cfd89959125416c0a21a6afcd2ffd384e ./jpg-0-0/00000004
.jpg

8e49a4423aa2415b530b43b5d8ab9dbe72e1f6578e65dddd6e4202938be380e1 ./jpg-0-0/00000005
.jpg

a8c87f31f3876d99110dd91a071697a424ac7f02aca4607941f7dacc26bc47f ./jpg-0-0/00000003
.jpg

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.jpg

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.jpg

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.jpg

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.jpg

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